Pervasive Computing Project Update

Team members

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Revised challenge statement

The challenge is to design a real-world version of the “Gandalf Staff” that can guide people in their environment by lighting the way forward. In the fantasy world, the staff works through magic, but in reality, we must create a framework and system that achieves similar guiding functions through technology. To keep the project feasible, the solution must balance ambition with the limits of budget and timeframe, and the problem can be viewed across three areas: hardware, software, and enclosure design, which together form the foundation for reimagining Gandalf’s staff in a modern context.

Revised Project Idea

Building on these challenges, our project is to create a “Gandalf Staff” reimagined as a smart walking companion. It blends safety, accessibility, and a touch of magic through sensors and voice interaction. The staff lights up automatically in the dark, with the glow symbolising Narya, the Ring of Fire entrusted to Gandalf. Inspired by its lore of hope and resilience, we reinterpret the staff as a symbol of guidance and support. It also responds to simple spoken commands, helping users navigate to a chosen location.

To achieve this, we will integrate:

- **Hardware**: NeoPixel Ring for main illumination, four LEDs for pointing the direction, a microphone and speaker for interactions, NEO-6 GPS module and MMC5603 magnetometer for navigation, and a microcontroller with a polymer lithium battery.
- **Software**: Logic for data handling and interaction, supported by Gemini API.
- **Enclosure**: A custom design resembling the original staff, built with 3D printing to balance aesthetics, durability, portability, and safety.

This reimagining captures the “magic” of Gandalf’s staff through technology, making it both a practical navigation aid and an inspiring companion.

User Journey

- **Context**: The user carries the staff as a navigation assistant.
- **Trigger**: The staff lights up automatically in the dark or responds to a spoken command such as “I want to explore a nearby monument”.
- **Steps**:
    - The staff glows to provide illumination.
    - The microphone captures the user’s voice command.
    - The system processes the speech, connects to a navigation service, and interprets the request.
    - Respective LEDs light up to give directions.
    - The user reaches the destination, and the staff return to standby.
- **After interaction**: Once the destination is reached, the staff returns to standby.

Inspiration sources

Our project evolved from a combination of imagination and technology. We were first inspired by The Lord of the Rings, where Gandalf’s staff represents guidance, light, and resilience. Both Tolkien’s books and the film trilogy gave us the idea of reimagining the staff in a modern, real-world form.

To make that vision possible, we explored research on wearable devices and navigation aids, which showed us how technology can support safety and accessibility. We also relied on technical tutorials and datasheets for components such as the NeoPixel Ring, GPS modules, and magnetometers. These resources helped us decide what was feasible and how we could bring our design to life.

By combining creative inspiration with academic studies and practical know-how, we shaped a project that captures the magic of Gandalf’s staff while staying grounded in real technology.

References (ACM style):

Jackson, P. (Director). 2001. *The Lord of the Rings: The Fellowship of the Ring*. New Line Cinema.
Jackson, P. (Director). 2002. *The Lord of the Rings: The Two Towers*. New Line Cinema.
Jackson, P. (Director). 2003. *The Lord of the Rings: The Return of the King*. New Line Cinema.
Tolkien, J. R. R. 1954. *The Fellowship of the Ring*. George Allen & Unwin.
Tolkien, J. R. R. 1954. *The Two Towers*. George Allen & Unwin.
Tolkien, J. R. R. 1955. *The Return of the King*. George Allen & Unwin.

Revised parts required

Hardware

- **Core system**: ESP32-E, Polymer Lithium Ion Battery 1100 mAh, connection cables, and Makerverse MicroSD Card Adapter.
- **Sensing and interaction**: Light sensor, microphone, speaker, and NEO-6 GPS module, MMC5603 magnetometer.
- **Lighting**: Neopixel Ring 16 LEDs for illumination.
- **Enclosure**: Rake handle for staff body with a 3D-printed case to house electronics.

Software

- **APIs and services**: Gemini API.
- **Design tools**: Fusion 360 and TinkerCAD for 3D design.

Revised Timeline

Task	**Who?**	**Week**									
		4	5	6	7	8	9	10	11	12	13
Research	All										
Finalise the idea concept	All										
Hardware Development	Ford										
Enclosure Printing	Ford										
Electrical Connection	Sydney										
Software Development	Kiba, Yu										
Documenting	All										
Component Integrations	All										
Prototype Testing	All										

Overall project progress

We’ve made solid progress so far. The electronic parts and rake handle are ready, and the project concept has been finalised with everyone on the same page about the use cases. The 3D model for the top part is about 80% done, with some sections already printed for team feedback.

The code structure is set up on GitHub, so we’ll kick off software development in Week 9 alongside the electrical connections. Hardware work is already underway, and enclosure printing is scheduled for the next couple of weeks. After that, we’ll move into documentation, integration, and finally prototype testing in Week 12.

Overall, things are running on schedule, and we’re about to shift from planning to full development.

Revised potential risks

Our team has more experience in software than in hardware or 3D printing, so we may need extra time to learn how to design, print, and assemble the enclosure. To manage this, we plan to start with simple prototypes and use online resources to build our skills early in the project.

Another risk is the availability of 3D printing and laser cutting facilities. If machines are busy or scheduling conflicts arise, the timeline could be affected. We will reduce this by booking equipment in advance, allowing buffer time, and preparing smaller test prints before the final design.

Finally, some hardware parts may not be available in labs or local stores, which could add cost and delay if we need to order them. Compatibility issues may also arise when combining sensors, microcontrollers, a power supply, and the enclosure. To prepare, we will identify alternative suppliers, keep backup components, and test modules individually before full integration. These risks are manageable and will help us grow as a team.

Individual contributions to date

Below are the contributions of each member to date.

Team Member 1: Kiba Lin

- Refining the project's core idea
- Project Proposal writing
- Rake handle selection and purchase

Team Member 2: Ford Pariyavatkul

- Refining the project's core idea
- Project Proposal and Project Proposal Update writings
- 3D enclosure development, printing and refinement
- Initial GitHub repository structuring and coding

Team Member 3: Yu Shen Qiu

- Refining the project's core idea
- Project Proposal writing
- Electronic BOM development and purchase

Team Member 4: Sydney Jingting Foo

- Refining the project's core idea
- Project Proposal writing
- Project management and control

Appendix

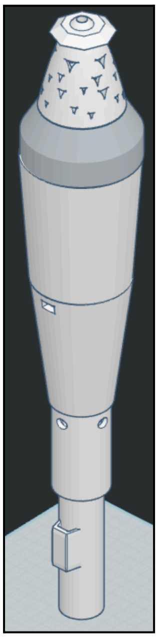

Fig 1. 3D Assembly of the top of the staff

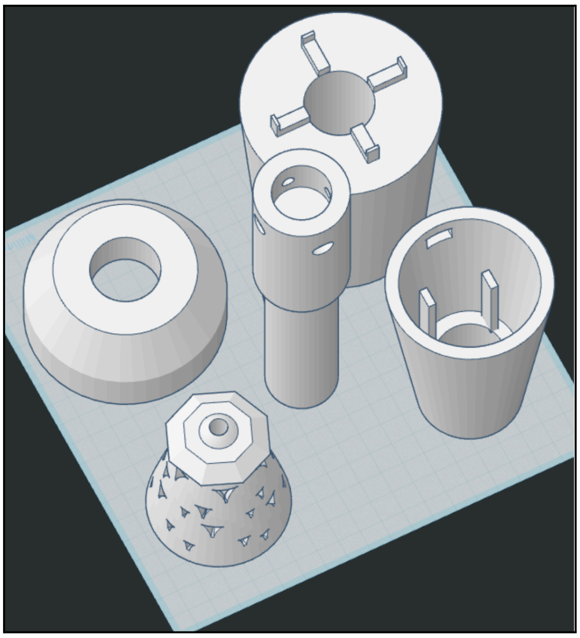

Fig 2. 3D parts of the top of the staff

Fig 3. The Gandalf Staff from the movie

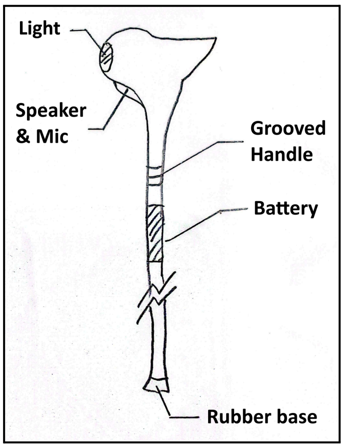

Fig 4. The Initial Sketch of the Proposal

Fig 5. The rake handle for the lower part of the staff